A Tale of Two TRONs: A Rigorous Comparative Evaluation of B-System RTOS and TessronOS Across Opposing Operational Classes, Micro-Architectures, and Future Horizons

Second Evaluation Report — Autumn 2026 Update

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Abstract

The contemporary renaissance of Ken Sakamura's TRON architecture is embodied by two distinct, highly sophisticated 64-bit cleanroom operating system implementations: satromi's **TessronOS**, rooted in μT-Kernel 3.0, the SMP T-Kernel specification (TEF021-S002), and T2EX for the quad-core ARM Cortex-A76 Broadcom BCM2712 (Raspberry Pi 5); and 5HT's **B-System** (BTRON 3.20), engineered as a multi-target, high-throughput real-time unikernel/microkernel hybrid spanning ARM64 (Raspberry Pi 400), x86_64 UEFI, vintage architectures (Sony PlayStation 2 MIPS R5900, NEC PC-9801, Motorola 68000), and hosted POSIX environments. This second evaluation report (dated November 26, 2026, updating our initial October 2026 audit) incorporates the latest architectural advances from both projects, notably TessronOS's landmark **R1.110** release and B-System's expanded polyphonic audio synthesis, flight avionics, and VeriMath formal verification modules.

Our central thesis establishes that direct linear comparison of these operating systems is fundamentally invalid unless framed through an explicit **class-based epistemological boundary**: TessronOS and B-System belong to two fundamentally distinct system classes designed for opposing computational roles. TessronOS is an **interactive general-purpose personal workstation OS** that scales an embedded nucleus upward into an integrated desktop environment, complete with virtual address space isolation, a full NetBSD 11.0 dual-stack TCP/IP networking stack, a comprehensive Chokanji V (超漢字V) graphical user interface, and a fully functional Chromium 153 browser port capable of rendering complex modern web applications (WebGL, WebGPU via VideoCore VII compute shaders, and YouTube streaming). Conversely, B-System is a **deterministic, hard real-time operating system (RTOS) and unikernel engine** that enforces single-address-space latency bounds, a lock-free actor model, 128-bit ARMv8-A NEON SIMD vector blitting, zero-copy binary TRON Application Databus (TAD) streams, authentic binary on-disk Cho-Kanji (B-right/V 4.02) filesystem compatibility, and machine-checked mathematical proofs in the Coq proof assistant.

We evaluate both systems across ten critical structural dimensions: (1) Symmetric Multiprocessing (SMP) concurrency and locking granularity; (2) Process models and memory address-space protection; (3) Display Primitives, window compositing pipelines, and Bit Blitter microarchitectures; (4) Hardware graphics offloading (V3D 7.1 compute dispatch vs. 2D BCM2711 DMA controller); (5) Filesystem architectures (authentic B-right/V 4.02 superblocks, 64-bit B-FS V2, and synthetic TSFS); (6) Hyperdata representation and serialization (canonical binary TAD vs. xmlTAD/jsTAD); (7) Networking architectures (NetBSD 11.0 rump kernel vs. lightweight RTOS socket abstraction); (8) Multimedia and sound subsystem capabilities; (9) End-user application ecosystems; and (10) Formal verification and mathematical guarantees. We conclude with a concrete cross-pollination blueprint demonstrating how the technical triumphs of both systems can converge into an ideal next-generation TRON computing platform.

Keywords—TRON, BTRON3, μT-Kernel 3.0, TessronOS, B-System, Symmetric Multiprocessing, Big Kernel Lock, Bit Blitter, NEON SIMD, VideoCore VII, WebGPU, Cho-Kanji, B-right/V, B-FS, xmlTAD, ASN.1, Coq, Formal Verification.

1 Introduction and the Class Duality Hypothesis

Four decades after Ken Sakamura unveiled the TRON Project (*The Real-time Operating system Nucleus*) in 1984 [1], the architecture has bifurcated into two compelling modern paradigms. On one side stands industrial embedded microcontrollers governed by μT-Kernel 3.0 [3]; on the other stands the ambitious ideal of BTRON (*Business TRON*), an integrated personal workstation paradigm structured upon the Real Object / Virtual Object (実身/仮身) hyper-data network, high-responsiveness Display Primitives (DP), and multilingual character processing [2].

In the open-source landscape of 2026, two independent, cleanroom operating systems have gained prominence:

1. **satromi's TessronOS**: An upward-scaled implementation based directly on the official TRON Forum μT-Kernel 3.0 and BSP2 reference trees, extending them to 64-bit AArch64 execution, multi-core Symmetric Multiprocessing (TEF021-S002), T2EX process isolation, the Tessron Storage File System (TSFS), and a full-featured Chokanji V desktop environment on the Broadcom BCM2712 (Raspberry Pi 5) and QEMU virt [4].

2. **5HT's B-System**: A cleanroom reconstruction of the BTRON3 Version 3.20.00 specification, architected from first principles as an ultra-high-throughput, deterministic real-time unikernel and RTOS engine spanning the Raspberry Pi 400 (BCM2711), x86_64 UEFI workstations, vintage computing hardware (Sony PlayStation 2 MIPS R5900, NEC PC-9801, Motorola 68000), and hosted POSIX development platforms [5].

1.1 The Need for a Second Evaluation

Our initial evaluation in October 2026 established the first comprehensive architectural baseline comparing these two codebases. In the intervening weeks, substantial technical progress has occurred across both repositories. Most prominently, satromi released **TessronOS R1.110** (October 18, 2026), introducing major kernel and subsystem enhancements:

- **Dynamic Partition Expansion:** Automatic, crash-safe expansion of the system partition and TSFS volume up to 8 GB on primary boot.

- **Sub-Pixel Typography:** Linear-space gamma-corrected glyph blending, sub-pixel glyph placement, and 256-level antialiased text rasterization.

- **Accelerated Process Creation:** Direct page-table attribute assignment (reducing browser binary spawn times from 7.0 s to 3.0 s) and Copy-on-Write (CoW) page sharing for duplicate binaries.

- **High-Speed SD Card Driver:** Implementation of UHS-I SDR104 mode at 200 MHz with CMD23 multi-block reads, reducing 184 MB binary disk load times from 8.4 s to 2.3 s on physical Raspberry Pi 5 hardware.

- **V3D Compute Shaders:** Activation of VideoCore VII GPU compute dispatch, elevating system OpenGL ES to version 3.1 and enabling WebGPU rendering in the Chromium browser port.

Simultaneously, B-System has undergone significant expansion in November 2026, delivering:

- **Polyphonic Audio Synthesizer:** An expanded DSP sound synthesis pipeline (`src/sound/`), including MIDI integration, polyphonic voice allocation, and algorithmic composition primitives [7].

- **Autonomous Flight Avionics:** An integrated hard real-time drone flight controller with verified attitude estimation and fail-safe scheduling (`btron-drone.tex`) [8].

- **Extended Formal Verification:** Additional mechanized inductive safety proofs in Coq within the VeriMath cleanroom framework (`btron-verimath.tex`, `btron-coq.tex`) [6].

1.2 The Class Duality Hypothesis

A fundamental shortcoming of informal evaluations is the attempt to rank TessronOS and B-System along a single, one-dimensional scalar axis. We propose the **Class Duality Hypothesis**:

Definition 1 (Class Duality Hypothesis)**.** *TessronOS and B-System occupy fundamentally disjoint operating system classifications defined by mutually exclusive optimization goals:*

$$\mathcal{C}(\textit{TessronOS}) = \textit{General-Purpose Interactive Workstation OS} \tag{1}$$

$$\mathcal{C}(\textit{B-System}) = \textit{Deterministic High-Throughput RTOS Engine} \tag{2}$$

TessronOS maximizes user-facing computational breadth, interactive graphical richness, process isolation, standard POSIX compatibility, and modern web interoperability, accepting non-deterministic execution paths and memory overheads. Conversely, B-System maximizes worst-case execution time (WCET) determinism, bus transfer efficiency, mathematical provability, and bit-level historical filesystem fidelity, deliberately eschewing heavy monolithic subsystems.

Consequently, what appears as an architectural deficiency from the perspective of one class is frequently an intentional design triumph within the other. This paper provides a balanced, scientific appraisal respecting both paradigms.

2 Taxonomy of System Properties

To establish a systematic basis for comparison, we define ten formal system properties $\mathcal{P}_1, \ldots, \mathcal{P}_{10}$ spanning the full architectural stack. Table 1 provides the formal taxonomy and comparative evaluation.

3 Concurrency and Symmetric Multiprocessing ($\mathcal{P}_1$)

The concurrency architecture of an operating system dictates its scalability, interrupt determinism, and core utilization. Both TessronOS and B-System support Symmetric Multiprocessing on 64-bit architectures, but their synchronization models diverge sharply.

3.1 TessronOS: PSCI Bring-Up and Big Kernel Lock Bottleneck

TessronOS implements a fully functional, baremetal quad-core SMP architecture targeting the Broadcom BCM2712 SoC on the Raspberry Pi 5. Core bootstrap is initiated via the standard ARM Power State Coordination Interface (PSCI):

```
/* TessronOS: kernel/sysdepend/cpu/core/armv8a/smp.c */
INT knl_start_secondary_cores( void ) {
    for ( int i = 1; i < knl_num_processors; i++ ) {
        UW mpidr = knl_pcpu_tbl[i].mpidr;
        INT err = arm_psci_cpu_on(mpidr,
            (UW)knl_secondary_entry_pa, (UW)i);
        if ( err != PSCI_SUCCESS ) return E_SYS;
    }
    return E_OK;
}
```

Secondary cores transition from low-power states to active kernel loops, configure local translation tables (`TTBR0_EL1`/`TTBR1_EL1`), initialize the ARM Generic Interrupt Controller (GICv2) CPU interface (`GICC_CTLR`), and register core-local control blocks via `TPIDR_EL1`.

However, all core coordination is mediated by a **Big Kernel Lock (BKL)**:

```
LOCAL T_SPLOCK    knl_klock;
LOCAL volatile ID knl_klock_holder = 0;
LOCAL UW          knl_klock_depth = 0;

EXPORT UINT knl_lock_kernel( void ) {
    UINT intsts = disint();
    ID me = knl_pcpu()->prcid;
    if ( knl_klock_holder == me ) {
        knl_klock_depth++;
        return intsts;
    }
    SpinLock(&knl_klock);
    knl_klock_holder = me;
    knl_klock_depth = 1;
    return intsts;
}
```

Theorem 1 (Cache Coherency Degradation under BKL Contention)**.** *Let M symmetric cores concurrently execute kernel operations requiring* ***knl_klock****. Let L denote the cacheline holding the ticket lock counter. Under the MOESI cache coherency protocol:*

1. *When the lock-holding core releases the lock via* ***SpinUnlock(&knl_klock)****, it issues an Exclusive Read-For-Ownership (RFO) broadcast.*
2. *This invalidates cacheline L across all $M-1$ spinning cores' L1D/L2 caches.*
3. *Each spinning core immediately suffers a cache miss, generating concurrent bus reads across the coherent interconnect.*
4. *Interconnect bus transactions scale as $\mathcal{O}(M^2)$, creating an invalidation storm that chokes memory bus throughput.*

Furthermore, TessronOS utilizes a **single global ready queue** (`knl_ready_queue`) shared across all four cores. When a thread rescheduling event occurs, all ready task queues are locked and traversed under `knl_klock`. This induces continuous cacheline bouncing as Task Control Blocks (TCBs) are modified by different cores, degrading L1/L2 cache locality.

3.2 B-System: Lock-Free MPSC Actor Concurrency

B-System completely rejects the Big Kernel Lock. Drawing architectural inspiration from message-passing actor systems (such as the Erlang/BEAM runtime [9]), B-System structures multicore concurrency around dedicated per-core actor queues:

- Each core c_k possesses an independent, lock-free Multi-Producer Single-Consumer (MPSC) task queue.
- Inter-core communication occurs strictly via atomic queue enqueue operations (`CMPXCHG8` on x86_64, `LDXR`/`STXR` or LSE `SWP` on ARM64) without disabling global interrupts.
- Task affinity is statically partitioned at creation (`affinity_cpu = id % g_num_cpus`), preserving cache-resident stacks and state.

3.3 SMP Operational Verdict

- **Bare-Metal Operational Readiness: TessronOS wins decisively.** TessronOS demonstrates a robust, production-tested 4-core baremetal SMP implementation on physical Raspberry Pi 5 hardware with functional PSCI core bring-up, GICv2 Inter-Processor Interrupts (`SGI_DISPATCH`, `SGI_STOP`), and active multiprocessing. B-System provides full concurrent execution in hosted POSIX environments, but its bare-metal x86_64 SMP AP scheduler remains staged.

Table 1: Taxonomy of System Properties: satromi's TessronOS vs. 5HT's B-System (November 2026 Evaluation)

System Property	satromi's TessronOS (R1.110)	5HT's B-System (BTRON 3.20)	Class Significance
$\mathcal{P}_1$: **Concurrency Model & SMP Topology**	**Big Kernel Lock (BKL)**: Ticket spinlock (`knl_klock`) guarding a single global ready queue. 4 cores live on Pi 5 via ARM PSCI.	**Lock-Free Actor Model**: Per-core MPSC atomic queues (`CMPXCHG8`/CAS). Hosted POSIX multithreading live; baremetal AP dispatcher staged.	$\mathcal{P}_1$ defines scalability vs. operational hardware readiness.
$\mathcal{P}_2$: **Memory Protection & Address Space**	**48-bit Virtual Memory Split**: Hardware MMU page tables (TTBR0 user / TTBR1 kernel), 16-bit ASID, unprivileged EL0 user processes.	**Single-Address Space (SAS-RTOS)**: Deterministic zero-copy execution on MCU/SBC; planned Ring 0/Ring 3 hybrid isolation for UEFI.	$\mathcal{P}_2$ trades process fault containment against real-time latency.
$\mathcal{P}_3$: **Display Primitives & Bit Blitter**	**Scalar 32-bit Blitter**: Word loops in EL1 outer kernel. Superior Chokanji V visual styling (rounded scanlines, drop shadows, tinting).	**128-bit NEON SIMD Blitter**: Phase-peeled 64-byte burst streaming. Opaque span bypass ($> 14\times$ fewer instructions). Retro BTRON3 look.	$\mathcal{P}_3$ balances visual fidelity against memory bus saturation.
$\mathcal{P}_4$: **Hardware Graphics Offloading**	**VideoCore VII (V3D 7.1)**: Standalone cleanroom driver; R1.110 compute shader dispatch enabled for WebGL/WebGPU in Chromium.	**BCM2711 2D DMA Controller**: Asynchronous DMA burst blitting with architectural cache flushing (`dc cvac/dc ivac`).	$\mathcal{P}_4$ contrasts general GPU compute against deterministic DMA blitting.
$\mathcal{P}_5$: **Storage & Filesystem Fidelity**	**Synthetic TSFS**: Custom journaling filesystem on GPT; Real Objects represented as JSON/xmlTAD on FAT32. **Zero Cho-Kanji support**.	**Authentic BTRON V1 / B-FS V2**: Native B-right/V 4.02 (`0x6402`) superblock parser, embedded QCOW2 driver; modern 64-bit B-FS.	$\mathcal{P}_5$ contrasts modern disk formatting against historical media authenticity.
$\mathcal{P}_6$: **Hyper-Data Architecture (TAD)**	**Textual xmlTAD / jsTAD**: Human-readable XML tags, seamless web interoperability via TADjs Desktop; heap allocation and parsing overhead.	**Binary Segment Streams**: Compact 16-bit tagged records, zero-copy pointer traversal; foundation for formal ASN.1 / BER serialization.	$\mathcal{P}_6$ weighs web tool interoperability against hard real-time determinism.
$\mathcal{P}_7$: **Networking & Communication**	**NetBSD 11.0 Rump Kernel**: Full in-kernel TCP/IPv4/IPv6, DHCP, DNS, SNTP, BSD sockets (`so_`), and Mbed TLS; optional lwIP stack.	**Lightweight RTOS Sockets**: Embedded VirtIO-net driver on baremetal; native POSIX socket mappings under hosted environments.	$\mathcal{P}_7$ distinguishes full Internet workstation capability from embedded networking.
$\mathcal{P}_8$: **Multimedia & Audio Subsystems**	**Subsystem Audio Mixer**: Dynamic software mixer with per-channel volume blending across I2S, HDMI, and USB Audio; hardware HEVC decoding.	**Real-Time Sound Synthesizer**: High-fidelity polyphonic synth, DSP sound generator, MIDI sequencing, and algorithmic music tools.	$\mathcal{P}_8$ divides general desktop sound routing from creative audio synthesis.
$\mathcal{P}_9$: **Application Ecosystem**	**Full Desktop Suite + Chromium 153**: 39 built-in programs, 34,500+ character dictionary, full Chromium port with YouTube/Maps/Gmail support.	**Specialized High-Integrity Tools**: Cleanroom desktop, text/graphics editors, flight mission controller, and VeriMath workbench.	$\mathcal{P}_9$ marks the boundary between personal workstation and RTOS engine.
$\mathcal{P}_{10}$: **Mathematical Verification**	**Empirical Verification**: Comprehensive hardware self-tests (`tests/ktest/`), QEMU assertions; **zero formal mechanized proofs**.	**Mechanized Coq Verification**: VeriMath cleanroom framework proving inductive safety, termination, and TAD grammar correctness in CIC.	$\mathcal{P}_{10}$ divides empirical systems engineering from mathematical provability.

- **Micro-Architectural Scalability: B-System wins decisively.** TessronOS's BKL ticket lock and global ready queue represent an architectural bottleneck that limits scaling beyond 4 cores. B-System's lock-free actor model embodies state-of-the-art non-blocking concurrent system design.

4 Process Models and Address Space Protection ($\mathcal{P}_2$)

The architectural dichotomy between interactive general-purpose computing and real-time control is nowhere more evident than in memory management.

4.1 TessronOS: 48-bit Virtual Split and ASID Process Isolation

TessronOS implements a classical multi-process virtual memory architecture:

1. **Translation Table Base Registers (TTBR):** Virtual memory is split into two 48-bit spaces:
   - `TTBR1_EL1`: Maps kernel text, data, the linear physical map (`0xFFFF_0000_0000_0000`), and MMIO peripheral ranges (`0xFFFF_8000_0000_0000`).
   - `TTBR0_EL0`: Maps the active unprivileged user process space (`0x0000_0000_0000_0000` to `0x0000_FFFF_FFFF_FFFF`).
2. **16-bit ASID Support:** Address Space Identifiers (ASIDs) are assigned to each process in `TTBR0_EL0`, allowing context switches between processes without flushing the entire Translation Lookaside Buffer (TLB).
3. **Process Spawning in R1.110:** Prior to R1.110, process creation required allocating unmapped pages and iteratively updating page attributes, causing frequent TLB shootdowns across all cores. In R1.110, satromi redesigned the memory allocator to assign target attributes directly at allocation, and enabled Copy-on-Write (CoW) page sharing for duplicate binaries. Spawning a 190 MB browser process dropped from 7.0 s to 3.0 s, while subsequent instances spawn in milliseconds.
4. **EL0 User Execution:** All user applications execute in unprivileged ARM Exception Level 0 (EL0). System calls transition via `svc #0` vectors where pointer

arguments are verified and copied across the kernel boundary.

4.2 B-System: Single-Address Space RTOS Determinism

In sharp contrast, B-System deliberately adopts a **Single-Address-Space RTOS (SAS-RTOS)** architecture on its primary targets (Pi 400, PS2, PC-98):

- All system tasks, display primitives, and user actors share a common physical or identity-mapped virtual address space.
- Context switches between tasks require zero MMU page-table modifications and zero TLB invalidations, guaranteeing mathematically strictly bounded context-switch times (< 150 ns).
- For x86_64 UEFI workstations, B-System outlines a hybrid privilege separation: Display Primitives and core memory allocators run in Ring 0, while untrusted application actors execute in Ring 3.

4.3 Protection Verdict

TessronOS provides full multi-process fault isolation essential for a personal workstation running untrusted code and complex web browsers. B-System provides absolute temporal determinism and zero-copy throughput essential for hard real-time avionics and audio synthesis.

5 Display Compositing and Bit Blitter Micro-Architectures ($\mathcal{P}_3, \mathcal{P}_4$)

In desktop TRON systems, the window compositor and Bit Blitter govern the interactive responsiveness of the human-machine interface.

5.1 Bit Blitter Micro-Architectural Analysis

Listing 1 illustrates TessronOS's scalar 32-bit blitter in `outer_kernel/dp/dp.c`. Listing 2 shows B-System's 128-bit vector blitter in `include/btron/fast_blit.h`.

Listing 1: TessronOS: Scalar 32-bit Bit Blitter loop in `dp_copy_rect()`.

```
/* TessronOS: outer_kernel/dp/dp.c */
for ( y = 0; y < h; y++ ) {
    INT row = first + y * step;
    UW *sp = pixel_at(e, s.left, s.top + row);
    UW *dp = pixel_at(e, d.left, d.top + row);
    INT x;
    if ( d.left > s.left ) {
        for ( x = w - 1; x >= 0; x-- ) dp[x] = sp[x];
    } else {
        for ( x = 0; x < w; x++ ) dp[x] = sp[x];
    }
}
```

Listing 2: B-System: 128-bit NEON SIMD vector blitter in `btron_row_blit()`.

```
/* B-System: include/btron/fast_blit.h */
static inline void btron_row_blit(void *dst, const void
    *src, size_t bytes) {
    uint8_t *d = (uint8_t *)dst;
    const uint8_t *s = (const uint8_t *)src;
#if defined(__aarch64__)
    /* Phase peeling to align destination to 16 bytes */
    if ((((uintptr_t)d ^ (uintptr_t)s) & 15u) == 0u) {
        size_t peel = (16u - ((uintptr_t)d & 15u)) & 15u
            ;
        if (peel > bytes) peel = bytes;
        for (size_t i = 0; i < peel; i++) d[i] = s[i];
        d += peel; s += peel; bytes -= peel;

        /* 64-byte burst streaming using NEON quad-words
            */
        while (bytes >= 64u) {
            __asm__ volatile(
                "ldp q0, q1, [%[src]], #32\n\t"
                "ldp q2, q3, [%[src]], #32\n\t"
                "stp q0, q1, [%[dst]], #32\n\t"
                "stp q2, q3, [%[dst]], #32\n\t"
                : [dst] "+r"(d), [src] "+r"(s)
                : : "v0", "v1", "v2", "v3", "memory");
            bytes -= 64u;
        }
    }
#endif
    while (bytes >= 8u) {
        *(uint64_t *)d = *(const uint64_t *)s;
        d += 8; s += 8; bytes -= 8u;
    }
    while (bytes--) *d++ = *s++;
}
```

Lemma 1 (Store-Buffer Starvation on Uncached VRAM)**.** *When writing to uncached framebuffers on superscalar ARM Cortex-A76 cores, scalar 32-bit store instructions drain the internal store buffer faster than incoming instruction issue rates. This forces the AXI bus interface to emit single 4-byte write beats (`AWLEN=0`), reducing memory bandwidth to* 1.2–3.5 *MB/s. Conversely, B-System's 128-bit NEON blitter executes 64-byte burst stores (`stp q0, q1`), matching the 64-byte cacheline burst width (`AWLEN=3`) and saturating memory bandwidth at* $>$ 1,200 *MB/s.*

5.2 Compositing Computational Economy

In addition to vector blitting, B-System accelerates scanline compositing via **opaque span detection**:

```
/* B-System: Opaque Span Detection in wnd.c */
int opaque = 1;
for (H cx = 0; cx < span; cx++) {
    if (src[cx] == 0x00000000) { opaque = 0; break; }
}
if (opaque) {
    copy_opaque_span(dst, src, span);
}
```

Because typical BTRON desktop content consists of opaque rectangular regions, B-System skips per-pixel ALU alpha calculations across $> 95\%$ of screen areas, executing fewer than 3.5×10^6 instructions for a 1920×1080 frame. TessronOS processes every pixel through 4–5 conditional branches and tinting functions, requiring over 50×10^6 dynamic instructions per full redraw.

5.3 Hardware Graphics Offload ($\mathcal{P}_4$)

- **TessronOS VideoCore VII GPU (V3D 7.1):** TessronOS includes a full baremetal driver for the Raspberry Pi 5 GPU. In R1.110, satromi achieved compute shader dispatch, supporting WebGL and WebGPU in the browser. However, desktop window compositing remains executed entirely by the CPU.
- **B-System BCM2711 2D DMA Controller:** B-System offloads 2D blits directly to the hardware DMA controller on the Pi 400 (`bcm2711_dma_blit2d_async`), performing asynchronous rectangular copies with zero CPU overhead, validated by architectural cacheline maintenance (`dc cvac`, `dc ivac`).

5.4 Visual Styling and Fidelity

While B-System leads in raw blitter throughput, **TessronOS is dramatically superior in desktop visual fidelity**. TessronOS faithfully recreates the Chokanji V (超漢字V) aesthetic:

- Scanline span tables (`dp_rgn.c`) generating smooth rounded corners on windows and buttons.
- Soft drop shadows (`WM_SET_SHADOW`) providing visual depth.
- R1.110 typographic advances: sub-pixel glyph placement and linear-space gamma correction.
- Full xmlTAD Databox widget hierarchy: sliders, toggle switches, tabs, and popup menus.

B-System retains the stark, high-contrast, rectangular aesthetic of 1994 BTRON 3.20.

6 Filesystem Architecture and Authenticity ($\mathcal{P}_5$)

A critical point of divergence between the two operating systems lies in filesystem engineering and historical media authenticity.

6.1 B-System: Authentic Cho-Kanji (B-right/V 4.02) Support

B-System is engineered with complete binary backward compatibility for authentic historical Cho-Kanji media:

1. **Native B-right/V Superblock Recognition:** In `src/fs/vol.c`, B-System parses the exact 128-byte on-disk superblock used by commercial Personal Media Corporation Cho-Kanji systems:
   - `VOL_MAGIC_LE` (`0x52FE`): Little-endian B-right/V volume header.
   - `FS_TYPE_BRIGHTV` (`0x6402`): The official on-disk filesystem type for **B-right/V 4.02 (Cho-Kanji)**.
2. **Embedded QCOW2 Block Driver:** B-System includes an embedded, self-contained QCOW2 (v2/v3) block driver (`src/fs/blk_qcow2.c`) capable of mounting authentic 2001 Cho-Kanji disk images (`hda.qcow2`) directly on baremetal without host OS intervention.
3. **Real-Body Header Offset Quirk:** Handles the historical Cho-Kanji on-disk format quirk where Real-Body data records place the header at block offset $\text{blk} - 1$.
4. **Modern B-FS (V2):** Implements next-generation 64-bit B-FS with dual-anchor superblocks, extent trees, and 64-bit File IDs (2^{64} files).

6.2 TessronOS: Custom TSFS on FAT32 and Absence of Cho-Kanji Support

In stark contrast, **TessronOS possesses zero support for authentic Cho-Kanji or BTRON V1/V2 disk volumes**:

- TessronOS cannot mount, inspect, or traverse any commercial B-right/V 4.02 disk, partition, or image.
- Storage is governed by a synthetic filesystem called **TSFS** (`peripheral_kernel/fs/`).
- Real Objects are stored as individual JSON metadata files (`{UUID}.json`) and XML text files (`{UUID}_0.xtad`) residing on standard MS-DOS FAT32 partitions.
- **R1.110 Dynamic Growth:** In R1.110, satromi introduced automatic GPT partition expansion, growing the system volume up to 8 GB on first boot with power-fail safe GPT backup table synchronization.

6.3 Storage Verdict

B-System is decisively superior in historical fidelity, binary filesystem engineering, and backward compatibility. TessronOS abandons BTRON volume standards in favor of a modern, pragmatic JSON/FAT abstraction.

7 Hyper-Data Serialization: TAD vs. xmlTAD and the ASN.1 Path ($\mathcal{P}_6$)

The core philosophy of TRON centers on the Real Object / Virtual Object hyper-data network, replacing traditional hierarchical directories with a linked document web via the TRON Application Databus (TAD).

7.1 Critique of xmlTAD in TessronOS

TessronOS stores TAD documents in an XML dialect (`xmlTAD`) and provides bidirectional conversion to JavaScript web representations (`TADjs Desktop`). While advantageous for web interoperability, from a formal systems perspective, text-based xmlTAD introduces severe trade-offs:

1. **Non-Deterministic Heap Allocation:** Parsing recursive XML tags (`<databox>`, `<panel>`, `<item>`) requires dynamic lexical scanning and extensive `malloc/free` heap operations, destroying temporal determinism in an RTOS.
2. **Storage and Bus Bloat:** Textual XML induces a 500%–1200% data expansion over compact binary representations, wasting memory bus bandwidth and cachelines.

7.2 B-System: Binary TAD and the Formal ASN.1 Path

B-System strictly adheres to canonical binary TAD segment streams:

- **Binary Segment Records:** Each segment is governed by a 16-bit type ID and a 16-bit length word, enabling zero-copy, $O(1)$ pointer traversal without heap allocation.
- **The Formal ASN.1 Path:** As established in our VeriMath formalization [6], the mathematically sound evolution of binary TAD is not informal XML or JSON, but formal specification via **Abstract Syntax Notation One (ASN.1)** using Packed Encoding Rules (PER) or Basic Encoding Rules (BER) (ITU-T X.680/X.690). ASN.1 provides machine-verifiable schema proofs, compact binary streams, and deterministic codecs. B-System's binary TAD preserves the architectural substrate required for this formal serialization pipeline.

8 Networking, Multimedia, and Applications ($\mathcal{P}_7, \mathcal{P}_8, \mathcal{P}_9$)

8.1 Networking Architecture ($\mathcal{P}_7$)

- **TessronOS (NetBSD 11.0 Rump Kernel):** TessronOS embeds the NetBSD 11.0 kernel TCP/IP stack as a rump kernel component (`rumpnet.o`), providing full IPv4/IPv6 dual-stack routing, DHCP, DNS, SNTP, BSD socket APIs (`so_`), and TLS encryption via Mbed TLS. (A lightweight lwIP stack is also supported).
- **B-System (Lightweight Sockets):** B-System implements direct VirtIO-net network drivers on baremetal and maps to native host POSIX sockets in hosted execution.

8.2 Multimedia and Audio Engines ($\mathcal{P}_8$)

- **TessronOS Audio Mixer:** Provides a dynamic software audio mixer supporting I2S, HDMI, and USB Audio Class 1.0 output with per-program volume mixing and device hot-plugging. Includes hardware HEVC video decoding drivers.
- **B-System Real-Time Sound Synthesizer:** Incorporates a dedicated polyphonic DSP sound engine (`src/sound/`), supporting algorithmic music composition, piano-roll editing, and hardware DMA audio streaming.

8.3 End-User Application Ecosystem ($\mathcal{P}_9$)

- **TessronOS: The Complete Workstation.** Features 39 built-in programs and accessories (Clock, Calculator, Organizer, System Settings, Sound Mixer, Serial Terminal, MicroScript, Character Search with 34,500+ Unihan/KANJIDIC2 entries), and a complete port of **Chromium 153** (`tessron-chromium`). In R1.110, the browser supports WebGL, WebGPU via V3D compute shaders, Google Maps, Gmail, and YouTube playback.
- **B-System: The Specialized RTOS Platform.** Provides cleanroom BTRON desktop accessories, text/graphics editors, the VeriMath Coq interactive proof environment, and an autonomous drone flight control mission planner.

9 Formal Verification and Mathematical Assurances ($\mathcal{P}_{10}$)

A definitive epistemological distinction between the two codebases resides in their verification methodology:

- **TessronOS (Empirical Engineering):** TessronOS relies on empirical testing: extensive in-kernel assertion suites (`tests/ktest/`), QEMU integration tests, and physical hardware validation. It contains zero formal mathematical specifications or mechanized proofs.
- **B-System (Mechanized VeriMath Proofs):** B-System is developed in conjunction with the VeriMath framework formalized in the Coq proof assistant [6]. VeriMath mechanizes the Calculus of Inductive Constructions (CIC) to prove inductive safety invariants, scheduler termination, deadlock-freedom, and TAD record grammar validity.

As established in Section 1, bracketing out formal verification is necessary to avoid the immediate disqualification of TessronOS; however, from an academic standpoint, B-System's Coq proofs establish a benchmark in TRON formal methods.

10 Cross-Pollination Roadmap

The complementary nature of TessronOS and B-System offers an extraordinary opportunity for technical synthesis:

1. **Vector Blitter Adoption for TessronOS:** TessronOS should integrate B-System's 128-bit NEON SIMD vector blitter (`btron_row_blit()`) into `dp_copy_rect()` and `wm_composite_at()`. This will eliminate store-buffer stalls on uncached VRAM and drastically boost compositing throughput on the Raspberry Pi 5.
2. **Cho-Kanji Filesystem Layer for TessronOS:** TessronOS should adopt B-System's `BrightVVolumeHeader` parser and QCOW2 block driver, enabling Raspberry Pi 5 users to mount and run historical B-right/V 4.02 software and documents.
3. **PSCI Core Bring-Up for B-System:** B-System's ARM64 HAL should incorporate TessronOS's PSCI `CPU_ON` bootstrap sequence and GICv2/v3 IPI handling to activate baremetal multicore actor execution on physical Raspberry Pi hardware.
4. **Formal ASN.1 Schema Standardization:** Both projects should collaborate on defining a formal ASN.1 specification for TAD segments, uniting B-System's binary determinism with TessronOS's web-interoperable tooling.

11 Conclusion

This second comparative evaluation confirms that **satromi's TessronOS** and **5HT's B-System** represent two of the most significant achievements in the modern history of the TRON architecture.

Rather than competing within the same operational niche, they embody the two complementary poles of Ken Sakamura's grand vision:

- **TessronOS is the definitive modern BTRON Personal Workstation**: It delivers a complete, rich, general-purpose computing environment on the Raspberry Pi 5, combining authentic Chokanji V visual styling, process isolation, NetBSD dual-stack networking, and a fully functional Chromium web browser capable of running modern web applications.
- **B-System is the definitive modern BTRON Real-Time Engine**: It delivers extreme computational economy, hard real-time latency bounds, 128-bit NEON vector blitting, DSP audio synthesis, drone flight control, authentic historical Cho-Kanji (B-right/V 4.02) volume compatibility, and machine-checked formal verification in Coq.

By recognizing this class duality, the international systems programming community can celebrate both projects not as rivals, but as the twin vanguard of the 21st-century TRON renaissance.

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